Modeling Knowledge

Yesterday

Today

Tomorrow

Slides

https://github.com/peter1328/presentations

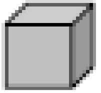
Peter Edelsbrunner
Turku 2026

Children develop spontaneous frameworks using their conceptual knowledge that are structured like a coherent theory!

You are both right and wrong! It differs between children – there is heterogeneity!

Stella, you are wrong! Children's knowledge is not theory-like, it consists of many loose intuitions!

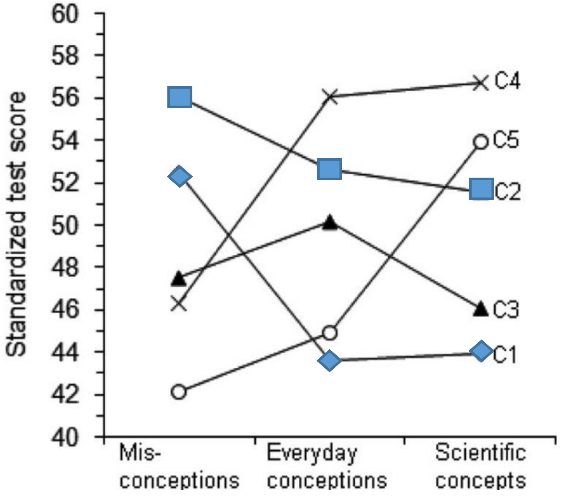
S.Vosniadou



M. Schneider

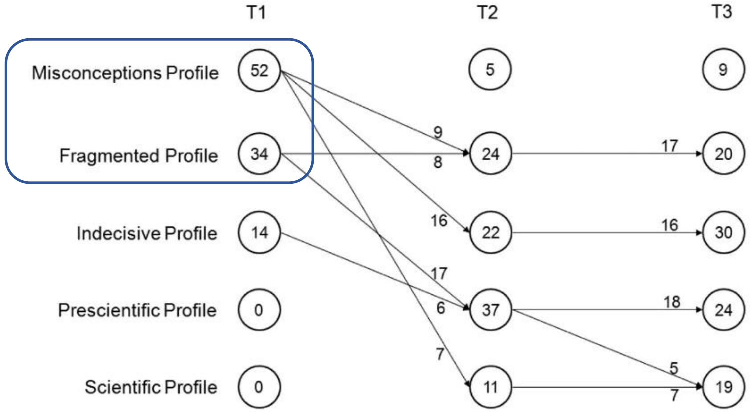
A. diSessa

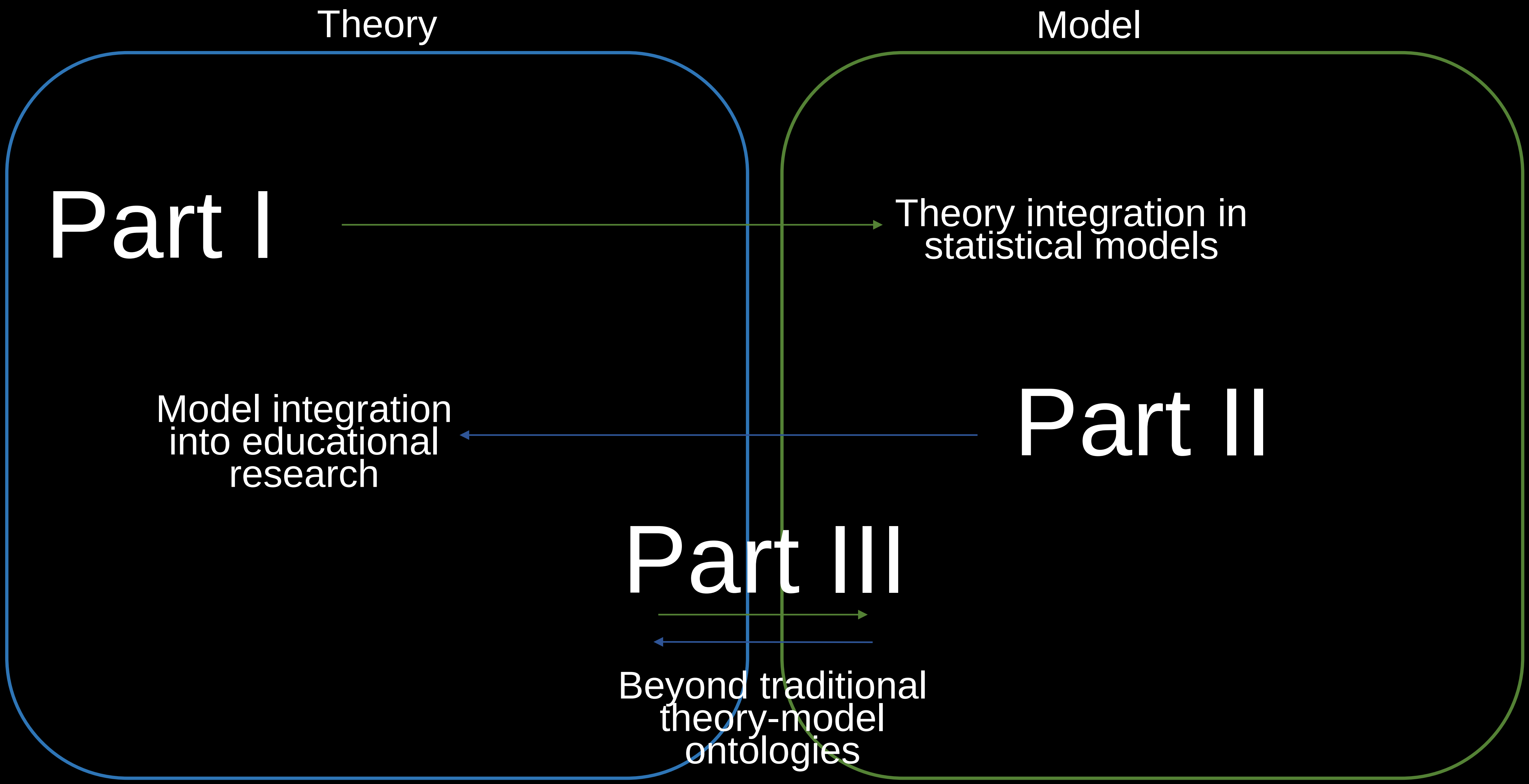
Schneider & Hardy (2013)
Latent transition analysis

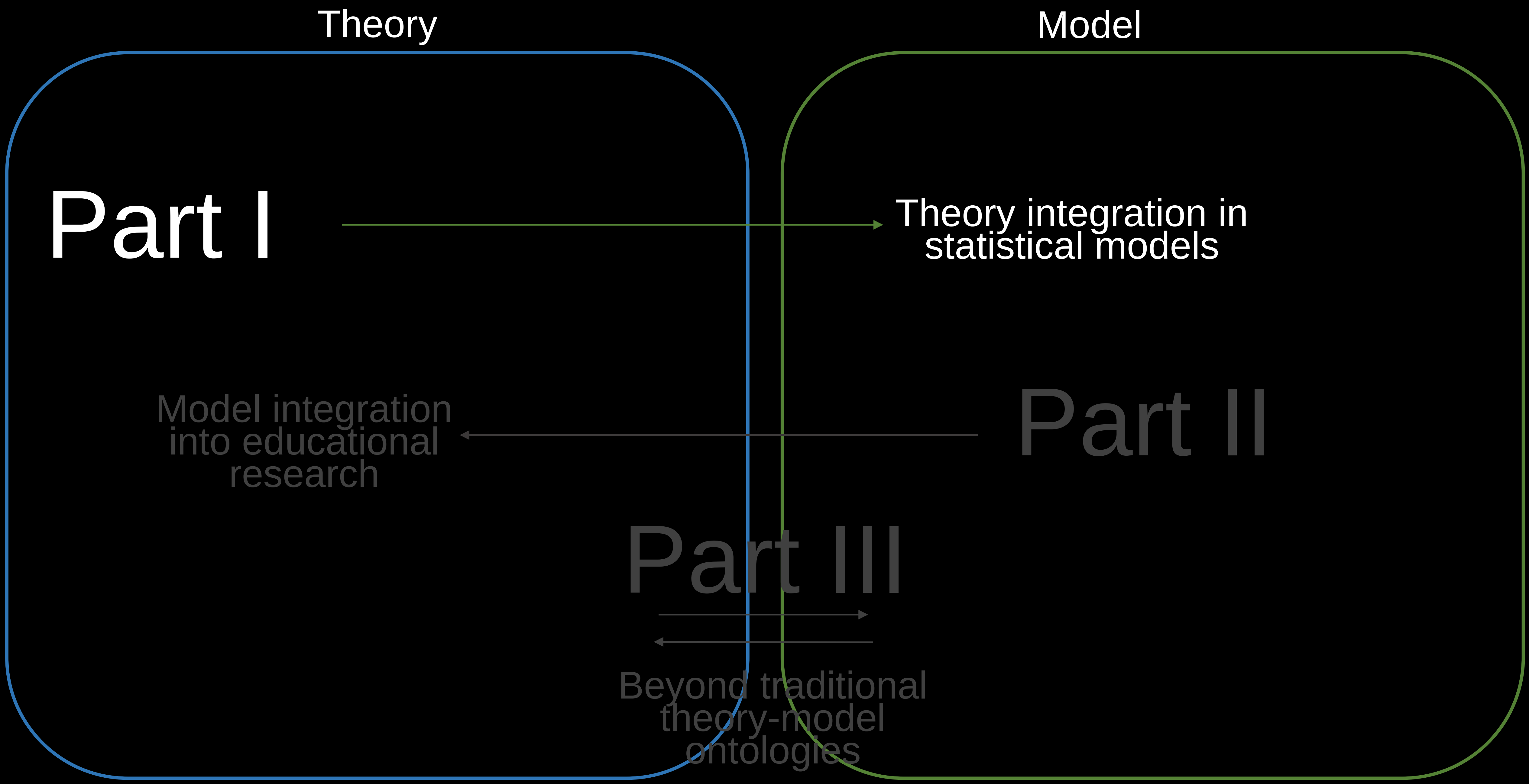


Mixture modeling

Children develop through knowledge states
(e.g., fragmented -> coherent)




Theory
Model
Part I
Theory integration in statistical models
Model integration into educational research
Part II
Part III
Beyond traditional theory-model ontologies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70s/80s

Theory

Inhelder & Piaget`s Theory of stage-wise development (1958)

Siegler`s (1976) ruled-based assessment methodology

Sato`s (1975) erroneous rules/bugs identification approach

Example subtraction items:

-16 - (-7)

2 - 11

12 + (-3)

Solution patterns:

000
001
010
100
011
101
110
111

Model

Kikumi Tatsuoka (1983)

Mapping:
Solution pattern <-> cognitive rule
(i.e., solution strategy)

e.g.
„011“ -> 80% prob. produced by misconception („bug“) leading you to subtract -7 from -16

Latent variable model (cf. *cognitive diagnosis model*):
- Bugs (i.e., misconceptions) as **latent mental states**
- Each response pattern produced by each bug with specific probability
- Emphasis on **patterns across multiple items**
- **Different strategies**/misconceptions can lead to **same** (sum) **score**

70s/80s

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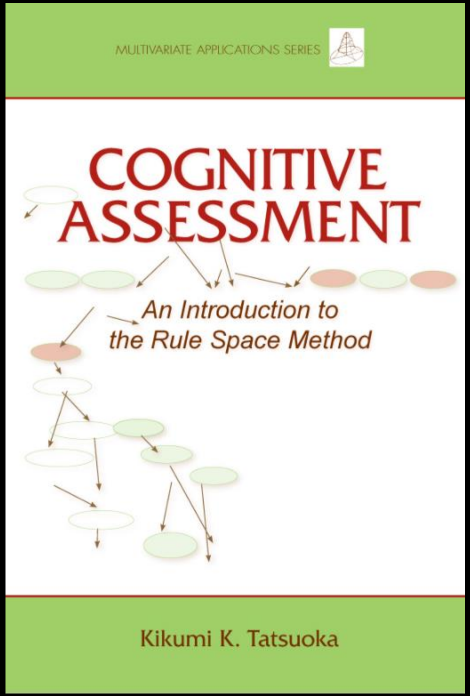
Model

Kikumi Tatsuoka (1983)

I`ve got an awesome idea!

Mapping:
Solution pattern <-> cognitive rule (i.e., solution strategy)



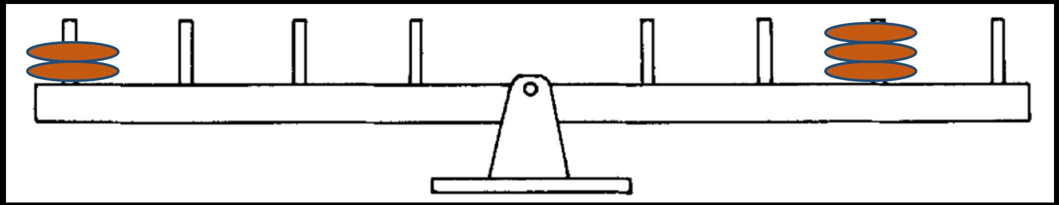
70s/80s

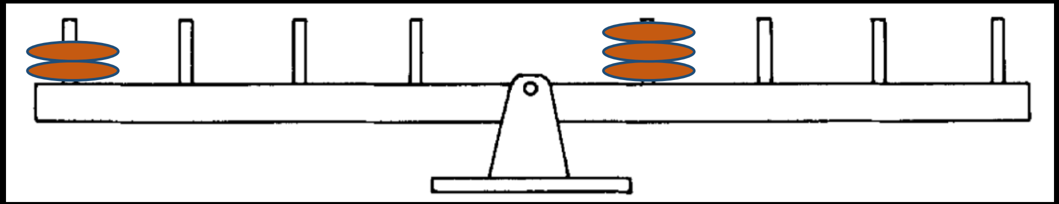
Theory

Inhelder & Piaget`s Theory of stage-wise development (1958)

Siegler`s (1976) ruled-based assessment methodology

Consider (1) weight, (2) distance, (3) proportion of w/d





Model

Mark Wilson (1985/1989)

This is so cool, let`s model it!

Extension of the **Rasch model**

Items have different difficulties but they depend on a person characteristic:
The cognitive rule (i.e., developmental stage)

Rule I (one-dimensional reasoning, e.g. weight)
…
Rule IV (proportional reasoning distance x weight)

Depending on rule person uses, specific items become easier (or harder)

70s/80s

Theory

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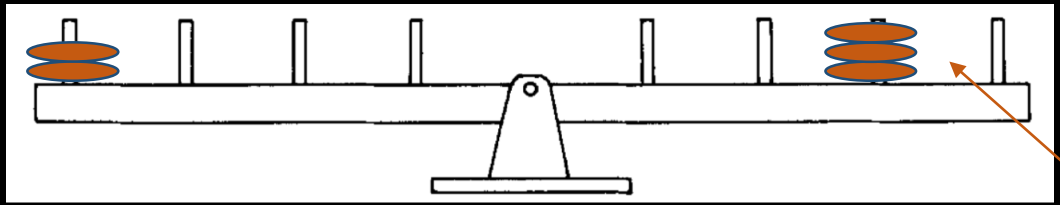
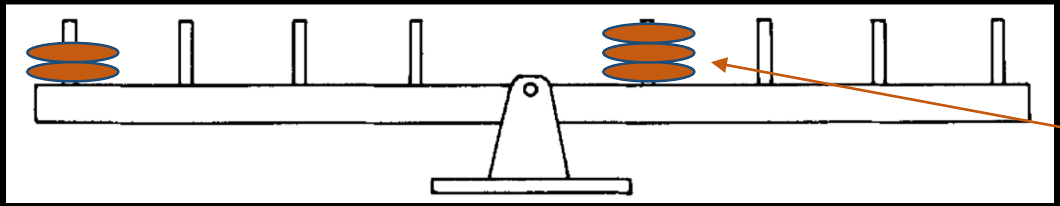
Siegler`s (1976) ruled-based assessment methodology

Consider (1) weight, (2) distance, (3) proportion of w/d





Model

Mark Wilson (1985/1989)

This is so cool, let`s model it!

Extension of the **Rasch model**

The **Saltus** model

Each **response pattern** indicates the **probability for** each of the four latent **mental states** (Rule I – IV)

Item types	RI	RII	RIII	RIV
Balance	1.00	1.00	1.00	1.00
Weight	1.00	1.00	1.00	1.00
Distance	.00	1.00	1.00	1.00
Conflict-balance	.00	.00	.33	1.00
Conflict-weight	1.00	1.00	.33	1.00
Conflict-distance	.00	.00	.33	1.00

$$\sigma_{ij} = \theta_i - \beta_j + \sum_h \phi_{ih} \tau_{hk}$$

70s/80s

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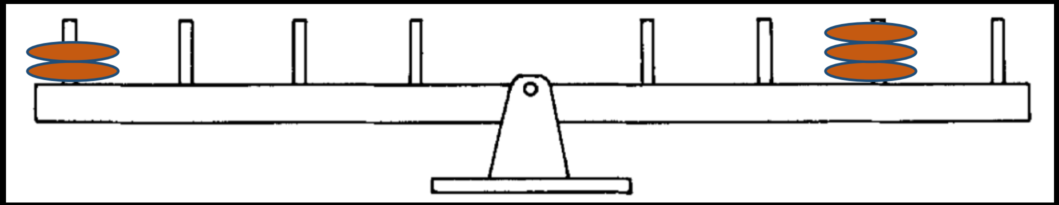
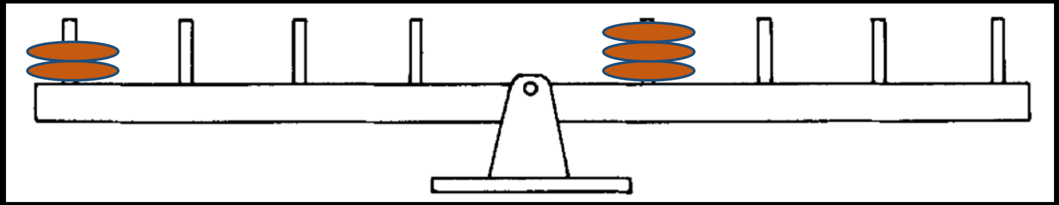
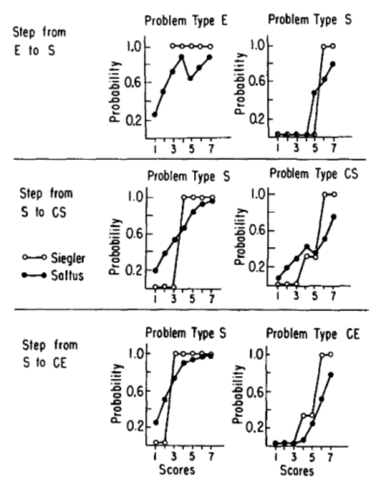




Model

Mark Wilson (1985/1989)



Model showed **deviations** from some of Siegler`s **predictions**

- Difference between person ability and item difficulty determines solution probability

- **Bonus/malus on solution probability** depending on latent rule state
- Latent rule *h* differs across persons, bonus/malus τ across item groups *k*

$$\sigma_{ij} = \boxed{\theta_i - \beta_j} + \boxed{\sum_h \phi_{ih} \tau_{hk}}$$

90s/2000s

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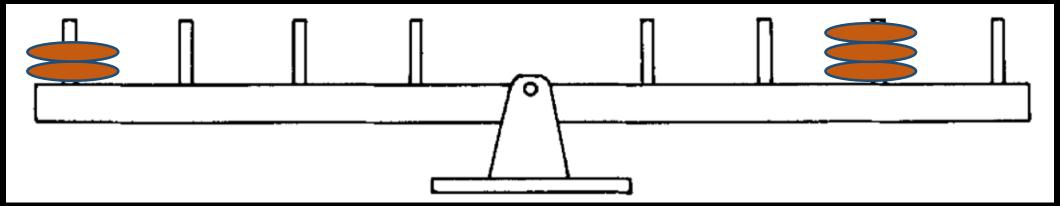
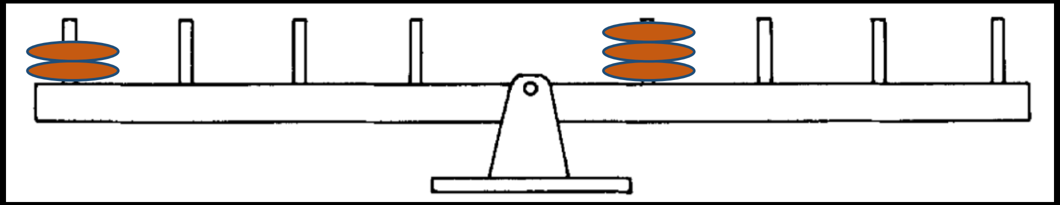




Model

Jansen & van der Maas (1997, 2002)

Constrained latent class analysis

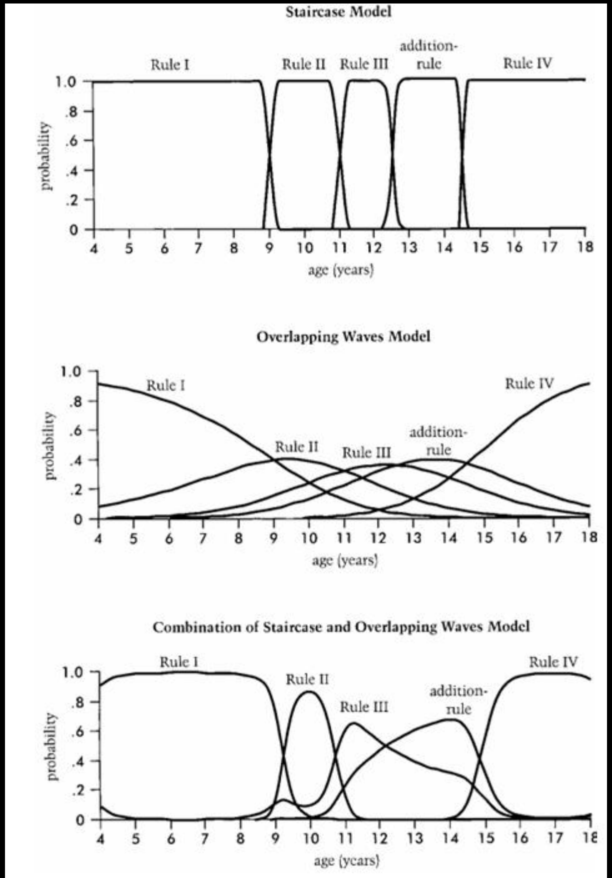
Formalized further potential rules:

- Random answer in case of balance
- Balance in case of w/d conflict
- Acknowledgment of *large* distances only
- Restriction of known item solution probabilities for Siegler Rules I – IV

90s/2000s

Theory

Result: Integrative theoretical *model of reasoning about the balance scale task*



Model

Jansen & van der Maas (1997, 2002)

Constrained latent class analysis

Results

N = 484 7th- and 8th-graders:
Rules I and II very consistent, rules above partially

Further applications:
- Rejection of connectonist models as rule-behaving
- Modeling of rules in transition phases, differences across age and development

Theory

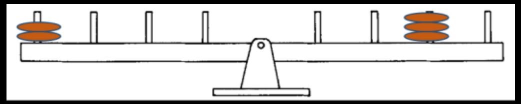
70s - 2000s

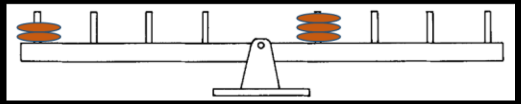
Model

Kikumi Tatsuoka (1983)

Mark Wilson (1985/1989)

Jansen & van der Maas (1997, 2002)





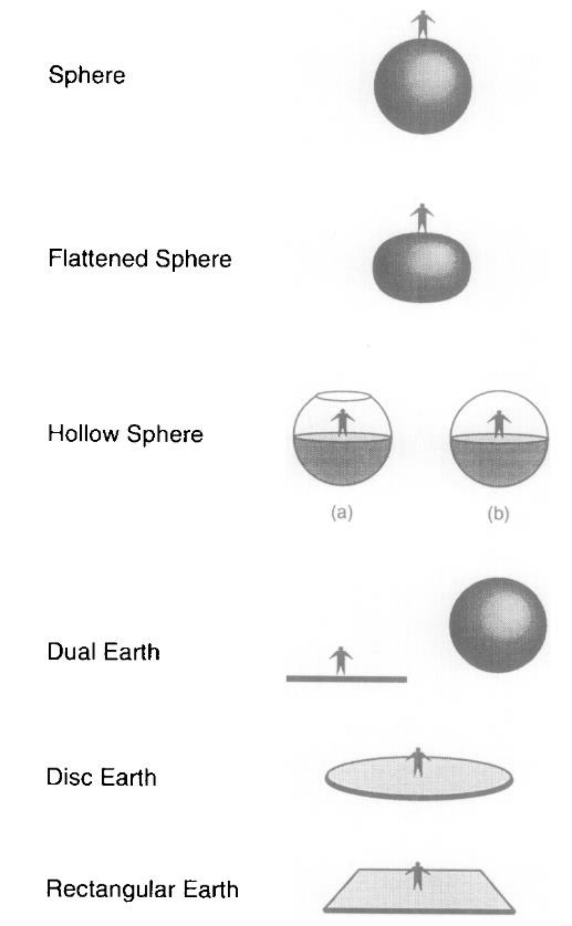
Modeling theoretical assumptions allowed:

- Determining **probabilistic** misconception/rule use for each child
- Estimating and comparing proportions of children showing each misconception/rule
- Estimating effects of practice/interventions on misconceptions/rule use
- Adaptive instruction (cf. Intelligent tutoring systems)
- Comparing theoretical predictions with empirical model estimates; theory development

Conceptual Change

Theory

Vosniadou & Brewer (1992)



Model

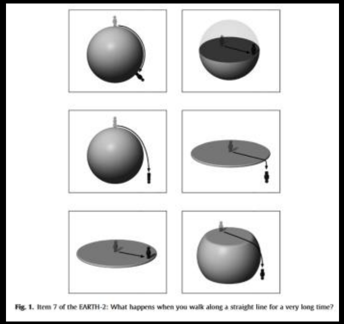
Straatemeier, Jansen, & van der Maas (2008)

- Children are typically classified according to answers in interviews, drawing, or claying tasks

- Arbitrary calssification cut-offs (e.g., > 80% in accordance with one predicted pattern), lack of possibility to identify unexpected patterns, no possibility to test goodness-of-fit (how well does our classification account for the observed data?)

- Constrained latent class analysis using a new pen-and-paper test (EARTH-2; 8 items + Interviews)

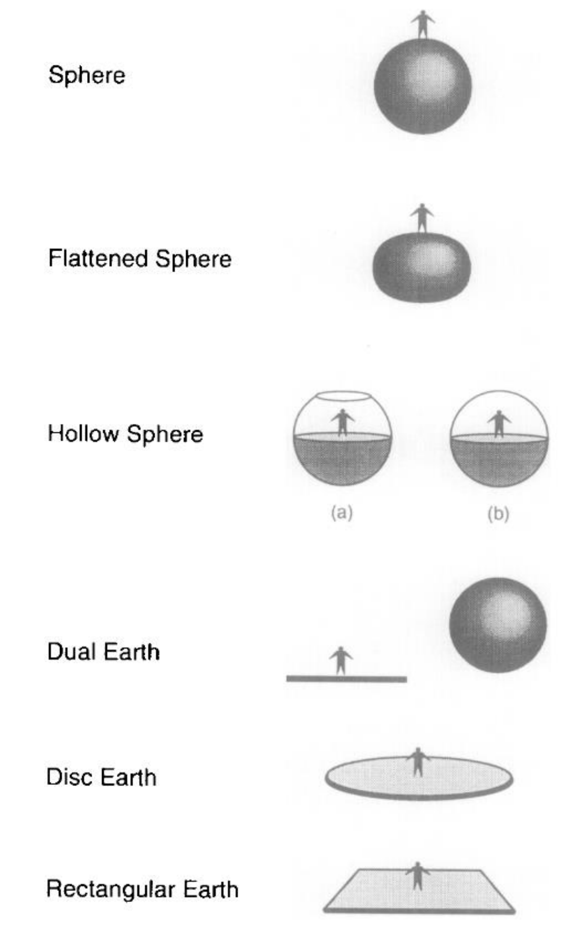
- Kindergarteners – 3rd grade



Conceptual Change

Theory

Vosniadou & Brewer (1992)

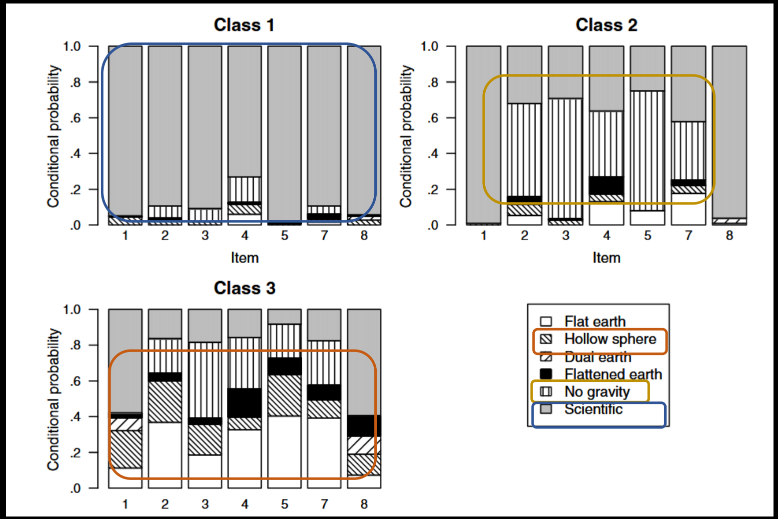


Model

Straatemeier, Jansen, & van der Maas (2008)

Results

Scientific concept



No gravity- conception

No gravity-/hollow sphere- conception

- Overall: Little evidence for coherent mental models, but also for fragmentation

Conceptual Change

Theory

Following up on Schneider & Hardy (2013)

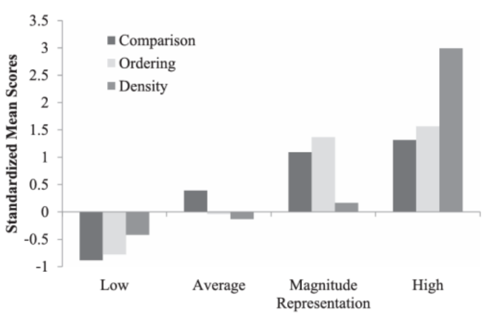
Theoretical assumptions about:

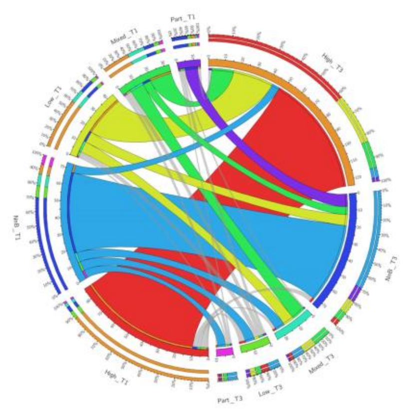
- Concepts about rational numbers
- Training hypothetico-deductive reasoning in children
- Higher education students` concepts of human memory

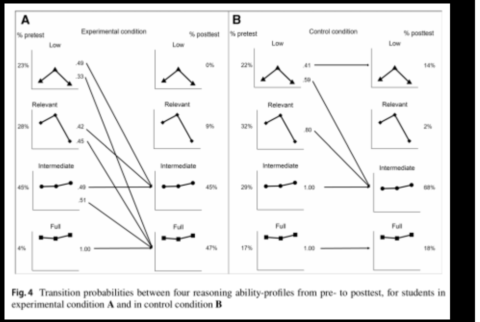
Model

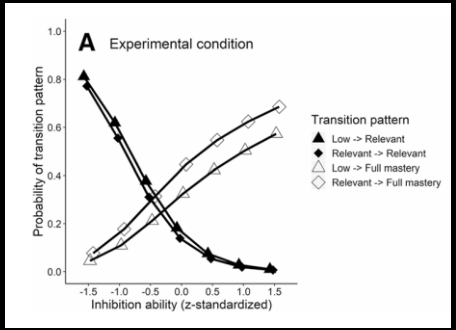
McMullen et al. (2015)
Edelsbrunner et al (2018)
Grimm et al. (2023)
Kainulainen et al. (2017)

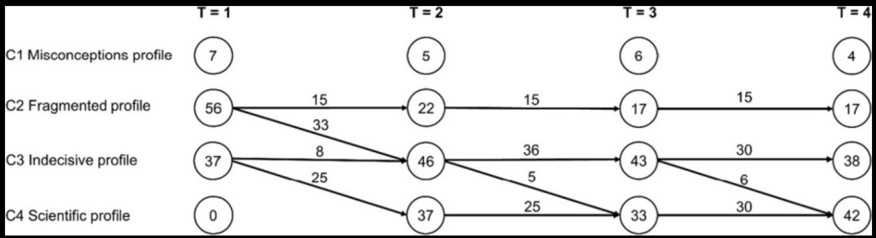
…











Conceptual Change

Theory

Model

Following up on Schneider & Hardy (2013)

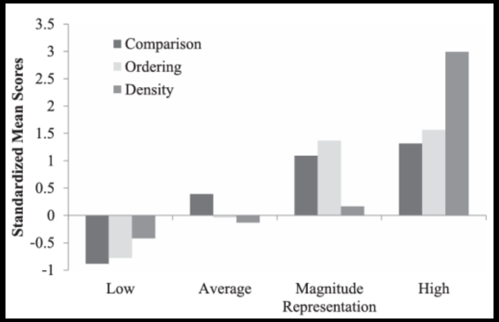
Novel theoretical insights:

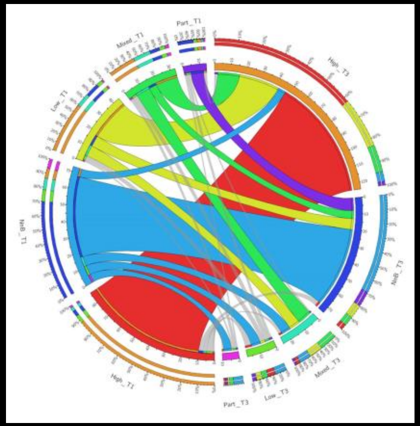
- Knowledge of magnitude representations is **necessary, but insufficient**, for knowledge of density concepts

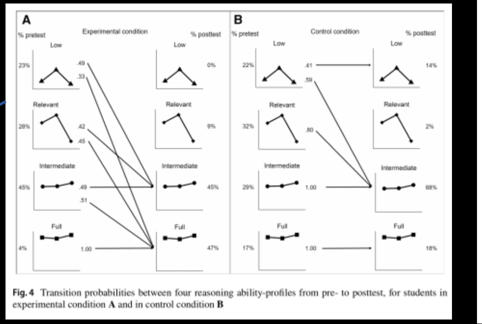
- **Inhibition** predicts acquisition of concepts about **insufficient evidence**

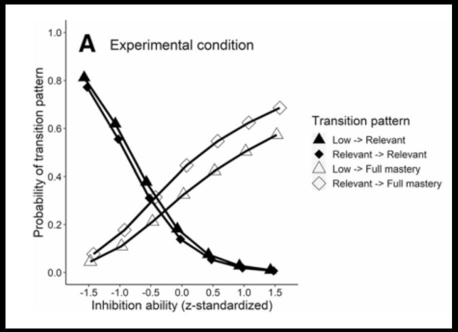
- **Knowledge integration** occurs during complex concept acquisition

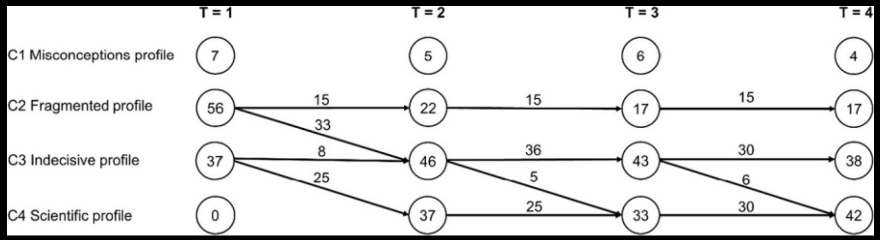
McMullen et al. (2015)
Edelsbrunner et al (2018)
Grimm et al. (2023)
Kainulainen et al. (2017)
…











Conceptual Change

Theory

Model

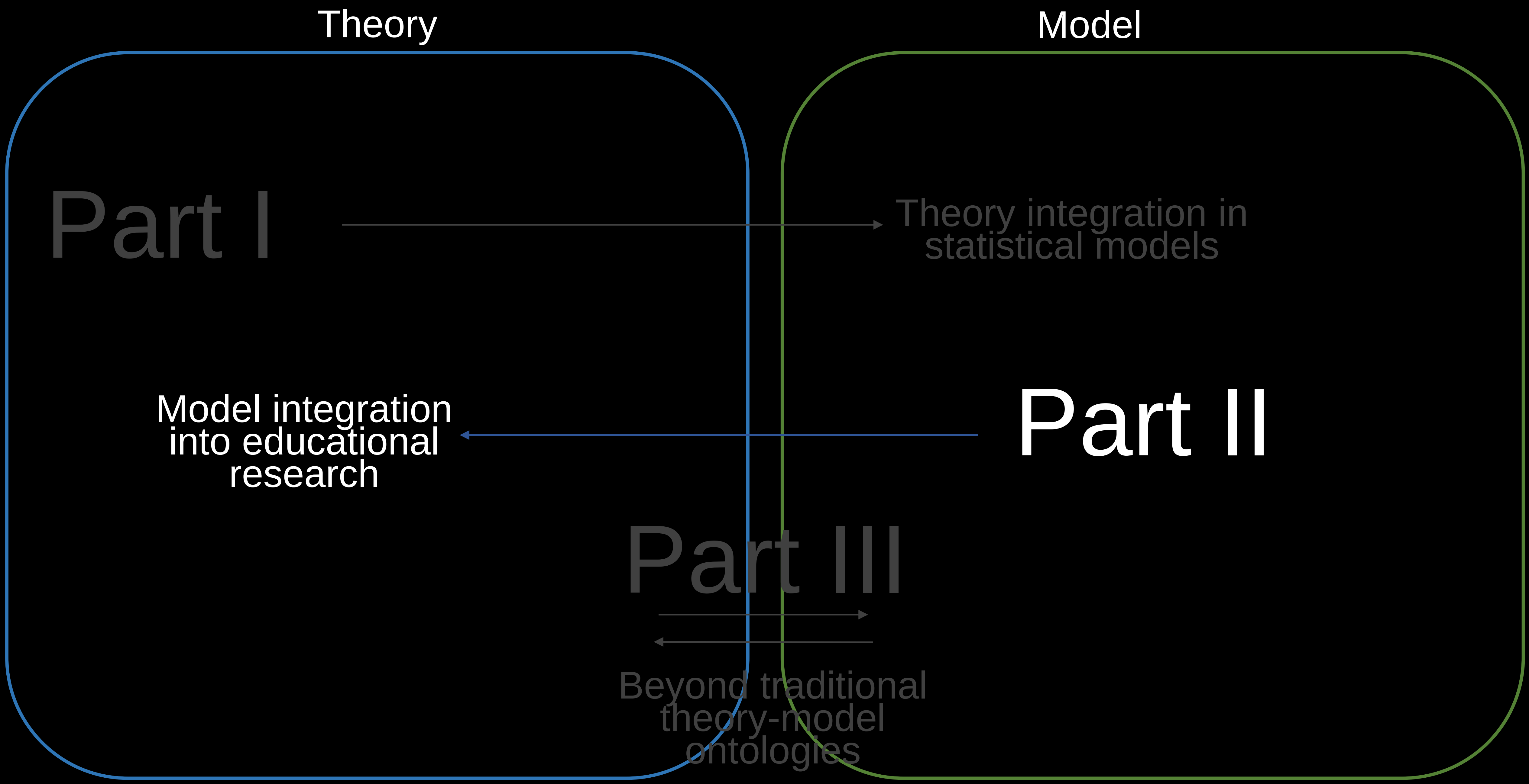
Historical observations:

- Move away **from theory-driven** (i.e., confirmatory) modeling **to data-driven** (i.e., exploratory) modeling

- Integrations of educational theory into statistical models centers around **qualitative individual differences** (e.g., stages, strategies, schemas, misconceptions)

- These individual differences are represented using **mixture modeling**

- Mixture models allowed **theory testing** and supported **theory refinement**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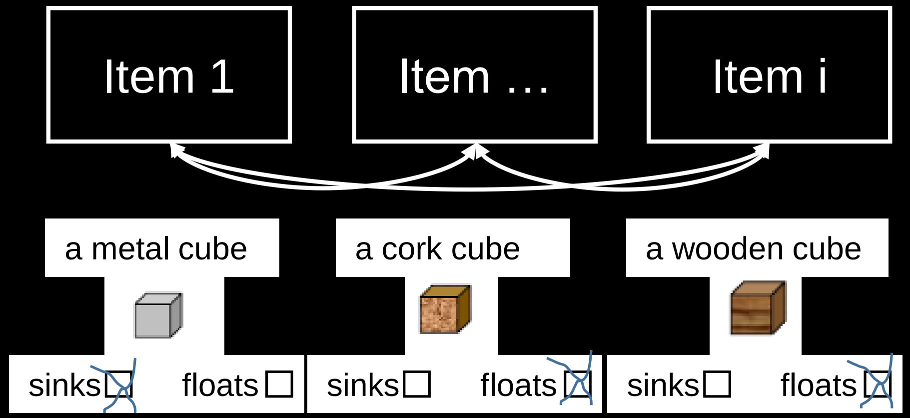
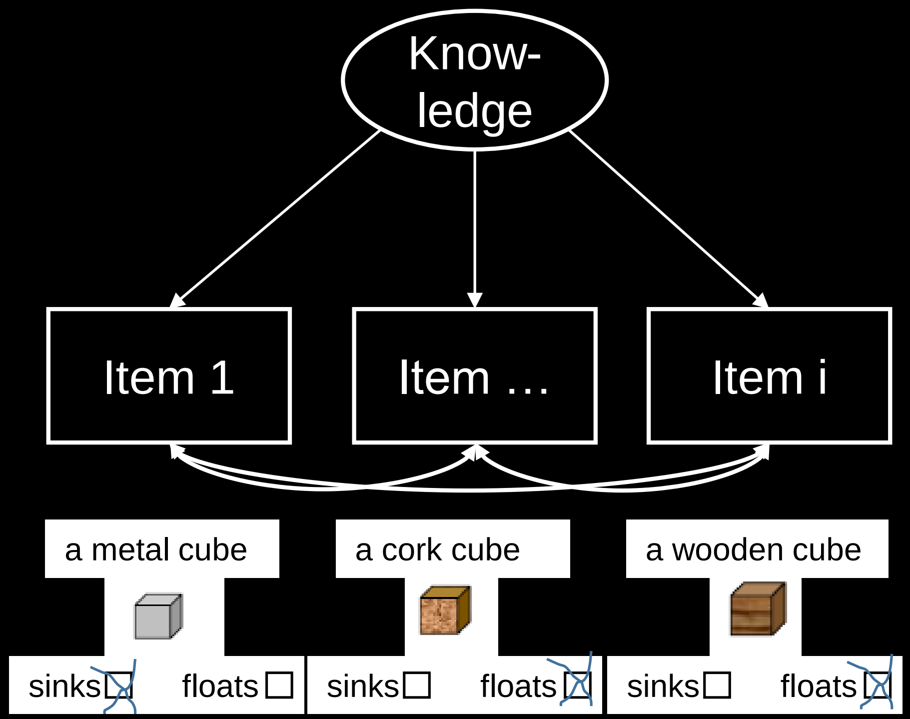


Responses to items **correlate** with one another

Theory

Model

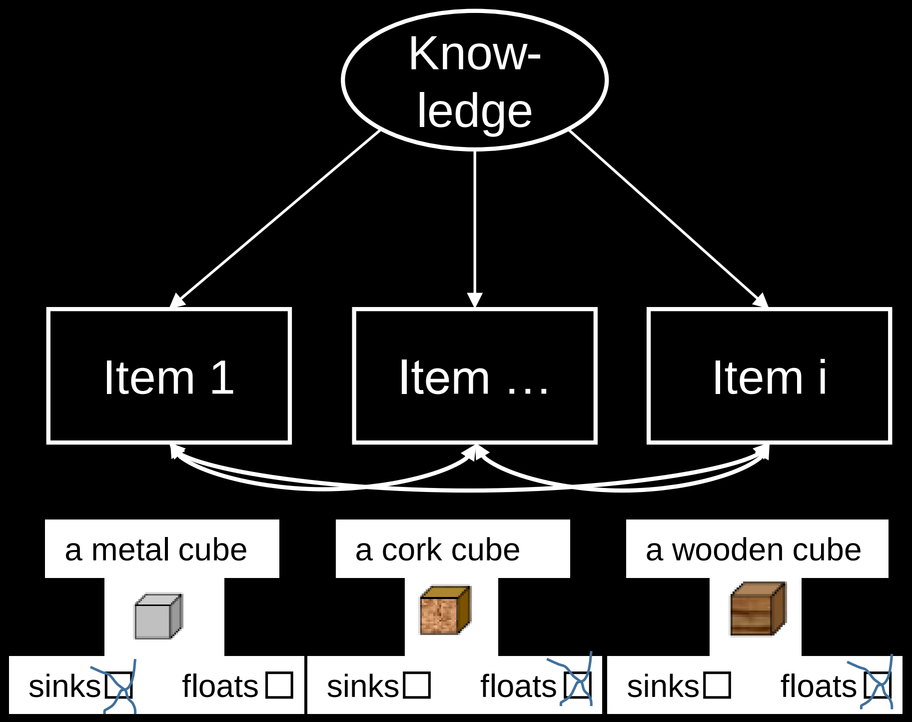


Responses to items **correlate** with one another

Because of a common underlying **factor** – Children`s **knowledge**

Theory

Model



Responses to items **correlate** with one another

Because of a common underlying **factor** – Children`s **knowledge**

Theory

If *we assume that all items measure knowledge about floating and sinking,* ***then*** *a factor model postulating one common factor must fit to our data*

90s: Ruling of **factor analysis ritual**

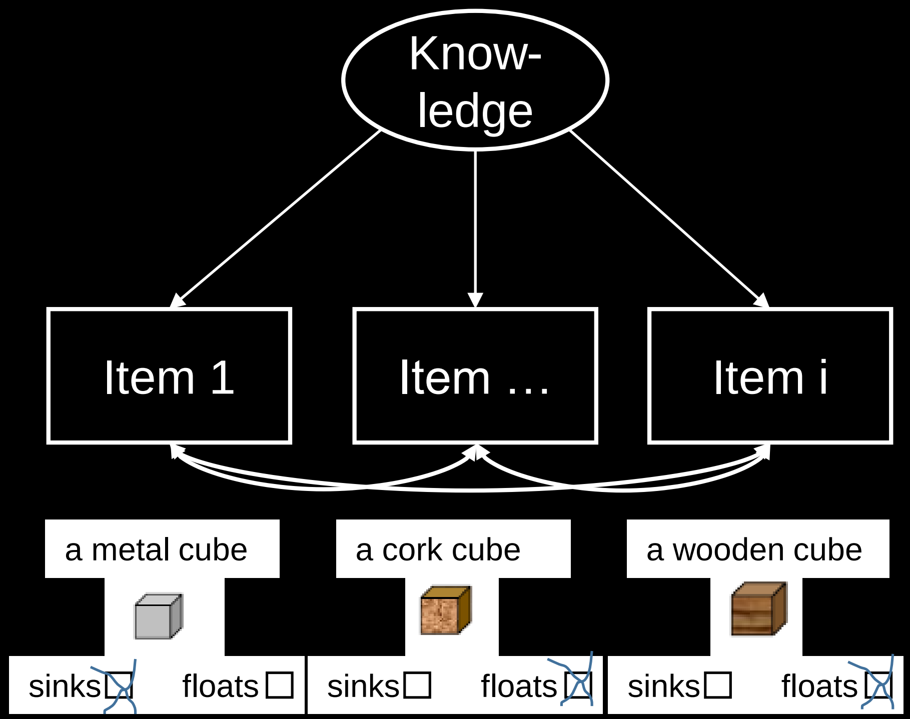
(1) Develop (many) items

(2) Fit a (exploratory or confirmatory) factor analysis model

(3) Compare results to theory

(4) Remove items that do not fit model

Model



Responses to items **correlate** with one another

Because of a common underlying **factor** – Children`s **knowledge**

Theory

Epistemic beliefs about knowledge, its sources, and development (Perry, 1970)

Beliefed to affect learning strategies and outcomes

90s:
Ruling of **factor analysis ritual**

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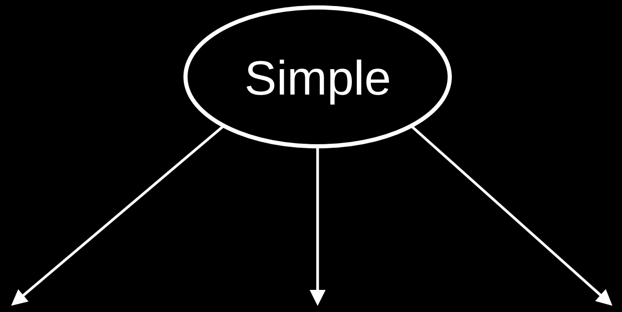
Model

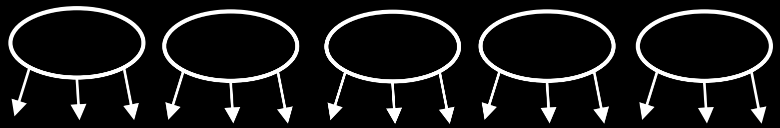
Schommer (1990, 1993; 1994)



Most words have one clear meaning

Being a good student generally involves memorising facts

Almost all information you can learn from a textbook you will get during the first reading



Source: Omniscient authority <-> reasoned out
Certainty: Absolute <-> evolving.
Organization: Compartmentalized <-> integrated
Control: determined <-> acquired
Speed: quick <-> gradual

Theory

Epistemic beliefs about knowledge and knowing (Perry, 1970)

Beliefed to affect learning strategies and outcomes

90s:
Ruling of **factor analysis ritual**

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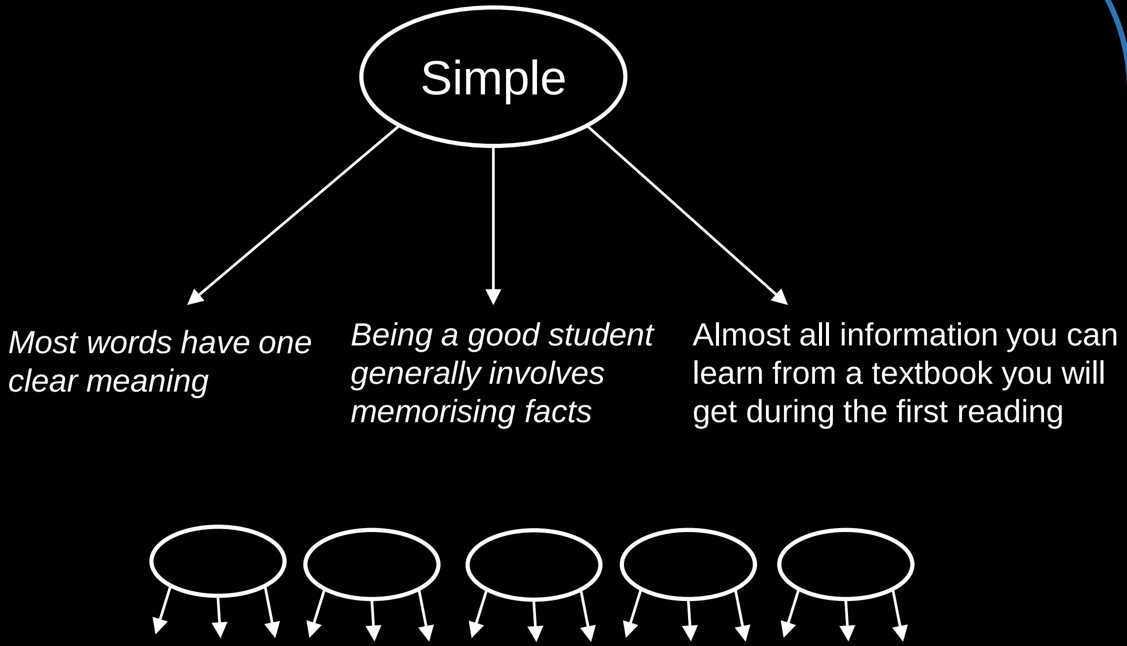
Schommer (1990, 1993; 1994)



Source: Omniscient authority <-> reasoned out
Certainty: Absolute <-> evolving.
Organization: Compartmentalized <-> integrated
Control: determined <-> acquired
Speed: quick <-> gradual

Theory

- Multiple new developments & revisions of epistemic belief questionnaires (Greene et al., 2010)
- Refinement of postulated dimensions and items (e.g., Chan & Elliott, 2004; Chai et al., 2006; Conley et al., 2004; Schiefer et al., 2022)

90s – 2000s: The **rise** of epistemic beliefs

(1) Develop (many) items

(2) Fit a (exploratory or confirmatory) factor analysis model

(3) Compare results to theory

(4) Remove items that do not fit model

Model

Simple

Most words have one clear meaning

Being a good student generally involves memorising facts

Almost all information you can learn from a textbook you will get during the first reading

- Move towards **epistemic cognition in action** because *you cannot measure what is not stable*

- If responses do **not correlate** between items, you should not regard them as a **unitary construct**

Item 1 | Item … | Item i

Theory

- Criticisms of instable factor structures across samples, low internal consistencies (Cronbach`s Alpha; DeBacker et al., 2008), contextual dependence (Merk et al., 2018), interpretational multiplicity (Greene & Seung, 2014)

- Handbook of Epistemic Cognition: **Nail in the coffin of epistemic beliefs research** (e.g., Mason, 2016)

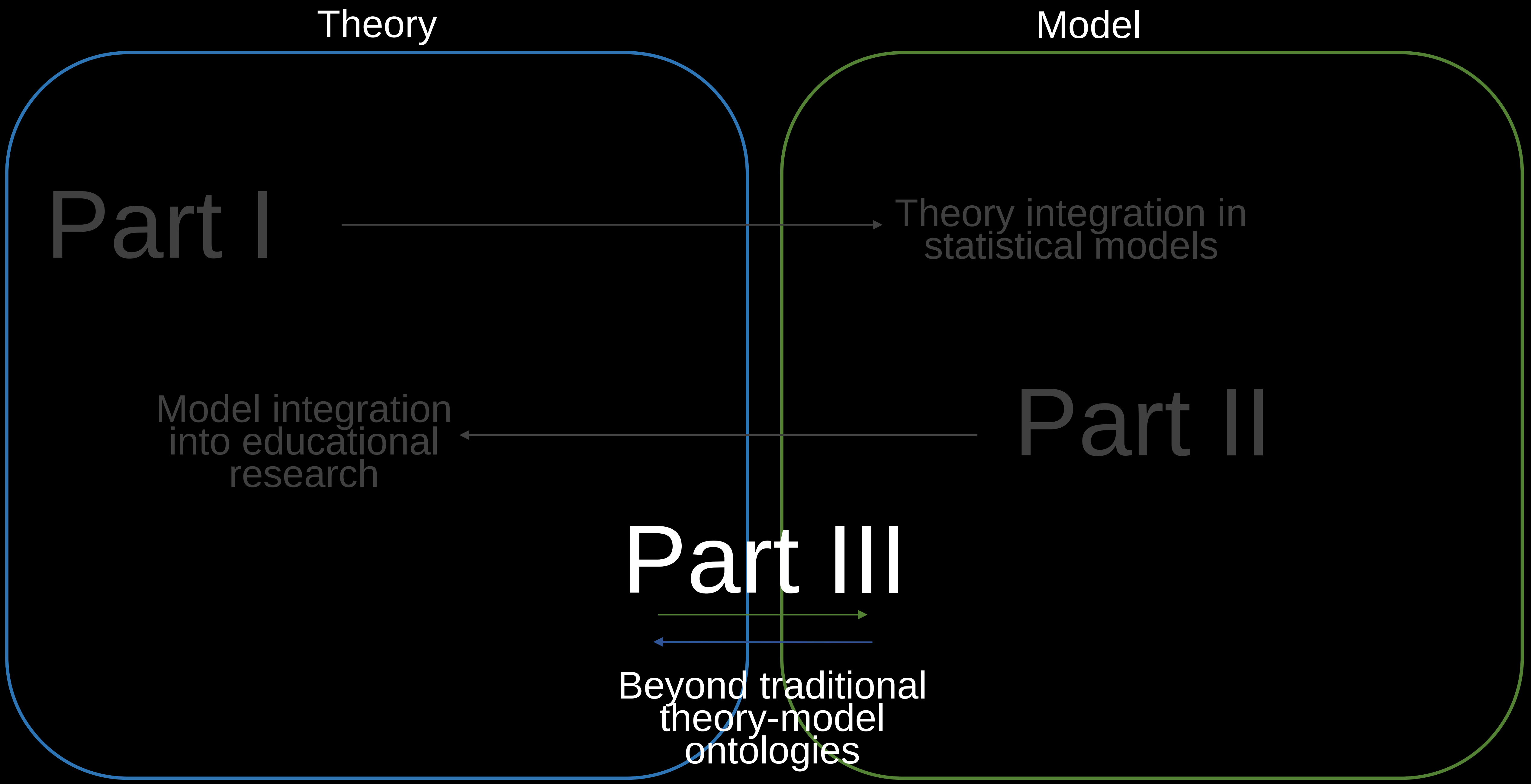
2010s: The **downfall** of epistemic beliefs

(1) Develop (many) items

(2) Fit a (exploratory or confirmatory) factor analysis model

(3) Compare results to theory

(4) Remove items that do not fit model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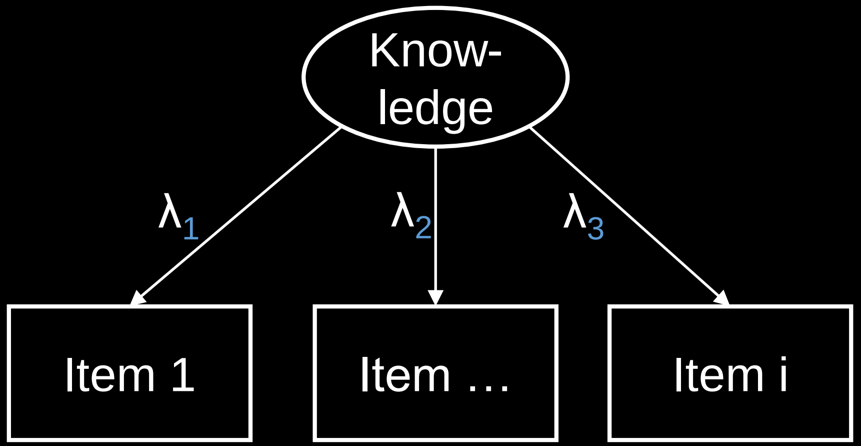


Since 1990s:

The factor structure (factor loadings etc.) must be the same across samples/ time points to be able to compare means, correlations, predictive strength (Meredith, 1993)

Measurement invariance

Theory

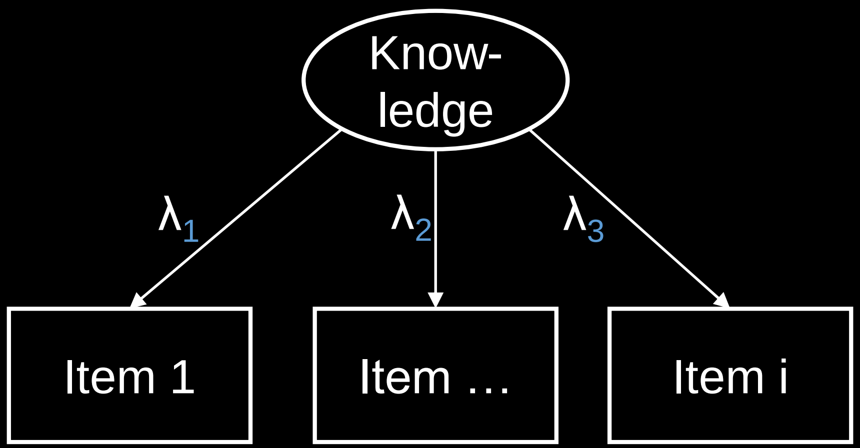
Since 2010s

Questionnaire/test data often thrown away if this statistical assumption is not given

Extreme case:
Epistemic beliefs (construct)

Wide-spread criticism of studies not investigating MI (Maassen et al., 2025)

Model



Since 1990s:

The factor structure (factor loadings etc.) must be the same across samples/ time points to be able to compare means, correlations, predictive strength (Meredith, 1993)

Measurement invariance

Theory

Alternative view

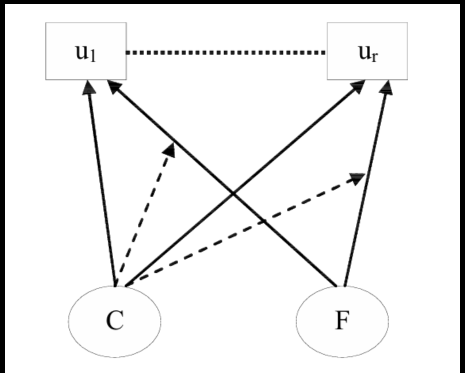
MI is **not a prerequisite for any comparisons** (Robitzsch & Lüdtke, 2023) because…

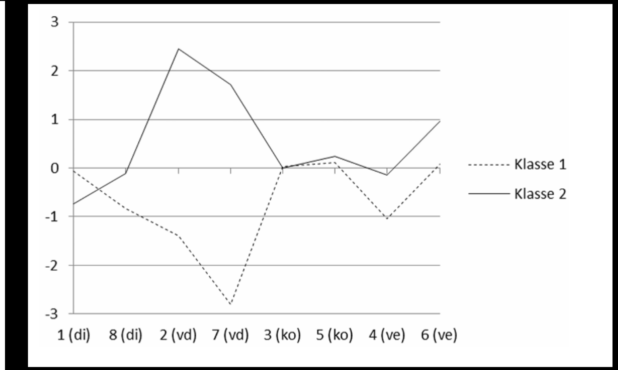
- MI modeling **decisions** are **arbitrary** (Robitzsch & Lüdtke, 2023)
- MI **models error** that we do not want in our model (cf. Boone & Staver, 2020)
- Analogy to moderation analysis:/causal inference A measurement model across groups is a **marginal/average** model

«Violation» of MI can be…

- Expected (cf. Conceptual change)
- Interesting

Model





Example: Kleickmann (2011)

Factor-mixture modeling

- Violation of MI becomes a **feature**
- Groups with measurement differences represent **knowledge restructuring**
- Children with same measurement properties regarding density, condensation, evaporation **differ in displamement** (cf. Wilson, 1993)

Theory

Alternative view

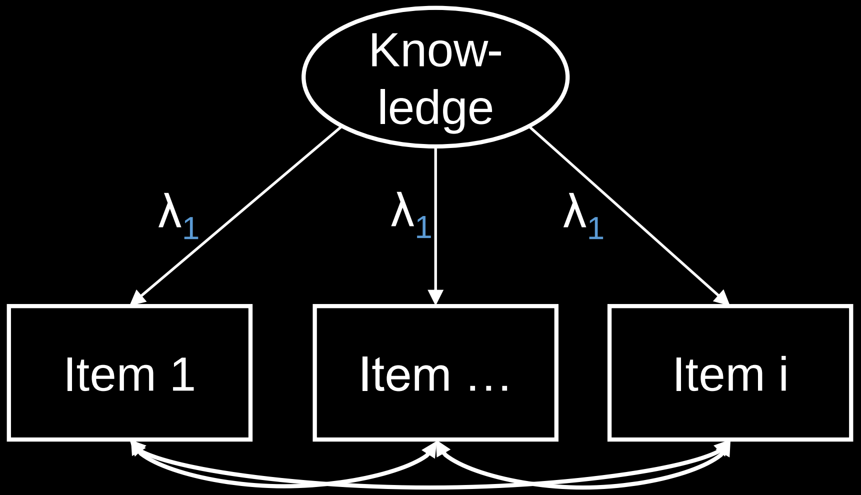
MI is **not a prerequisite for any comparisons** (Robitzsch & Lüdtke, 2023) because…

- MI modeling **decisions** are **arbitrary** (Robitzsch & Lüdtke, 2023)
- MI **models error** that we do not want in our model (cf. Boone & Staver, 2020)
- Analogy to moderation analysis:/causal inference A measurement model across groups is a **marginal/average** model

«Violation» of MI can be…

- Expected (cf. Conceptual change)
- Interesting

Model



The sum score is a *constrained version* of factor analysis
(McNeish & Wolf, 2020)

- All items are *equally weighted*

- Represents the implicit assumption of *equal factor loardings*

because

Theory

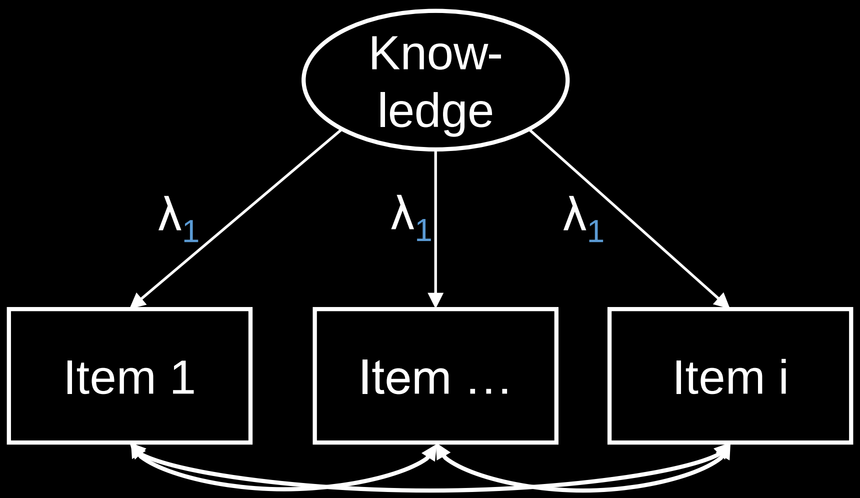
nope! → 0. Sum/mean scores make the assumption of equal factor loadings

OFC! → 1. This assumption is testable via factor analysis

Yes! → 2. Empirically, this assumption rarely holds

nope! → 3. Conclusion:
We should not use sum scores/means at all

Model



Theory

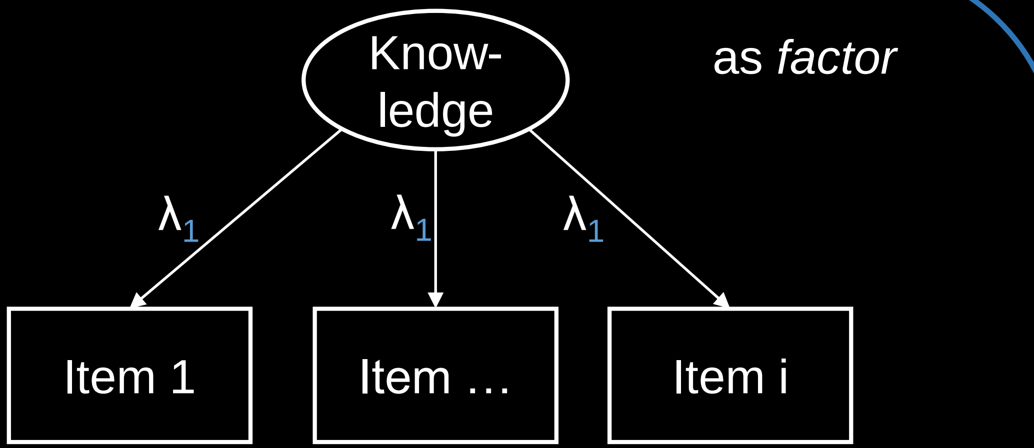
0. Sum/mean scores make the assumption of equal factor loadings

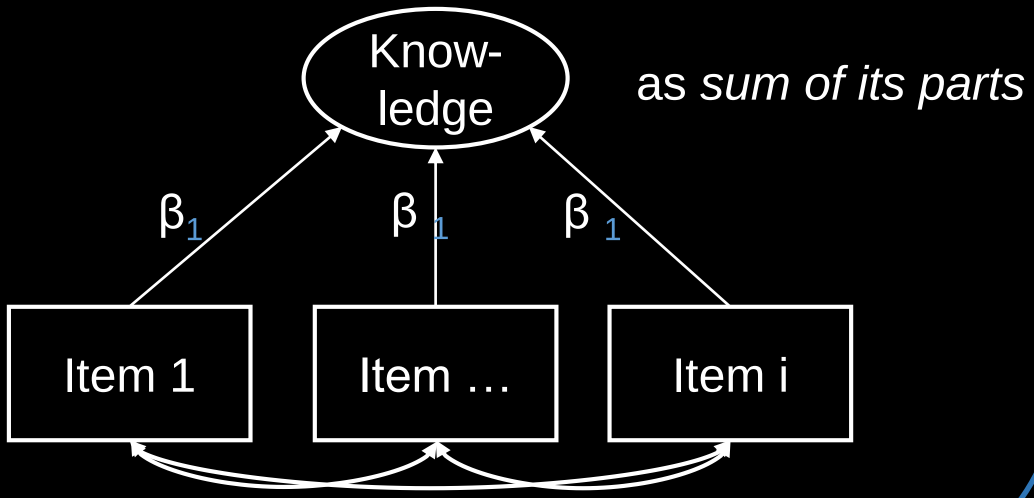
1. This assumption is testable via factor analysis

2. Empirically, this assumption rarely holds

3. Conclusion:
We should not use sum scores/means at all

Model





Theory

0. Sum/mean scores make the assumption of equal factor loadings

1. This assumption is testable via factor analysis

2. Empirically, this assumption rarely holds

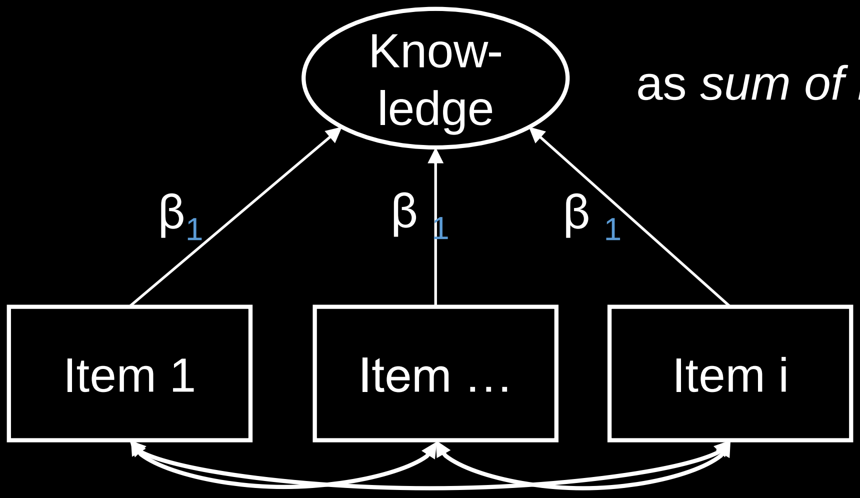
3. Conclusion:
We should not use sum scores/means at all

Model

Knowledge can be considered a **formative** or **composite** construct (Taber, 2018; Stadler et al., 2021; Edelsbrunner, 2022; Edelsbrunner et al., 2025)

We can use it as a **functionally unidimensional** construct (Baird et al., 2017)

No need for interitem correlations or high Cronbach`s Alphas (Edelsbrunner et al., 2025)



as *sum of its parts*

Theory

0. Sum/mean scores make the assumption of equal factor loadings

1. This assumption is testable via factor analysis

2. Empirically, this assumption rarely holds

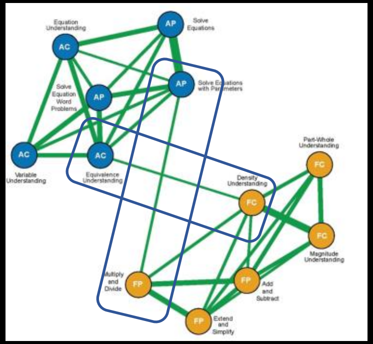
3. Conclusion:
We should not use sum scores/means at all

Model

- **Mutual** dependencies/ developmental effects between items (van der Maas et al., 2006; vanderWeele & Batty, 2023)

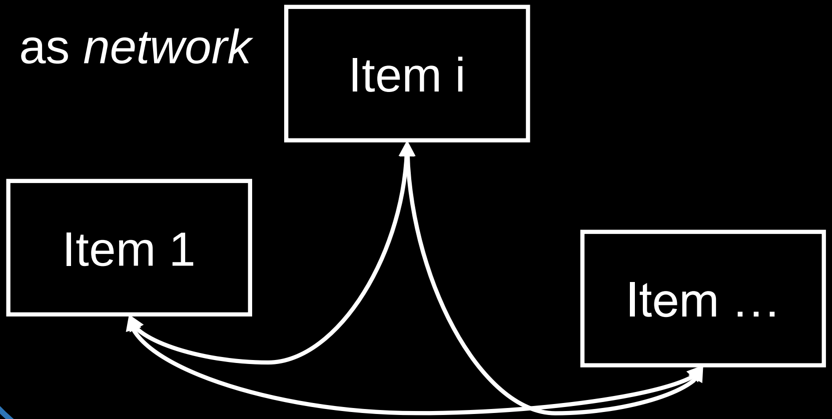
- **No** latent variable

- Unique (i.e., partial) correlations between items point to **mutual dependencies**/influences



Example: Forsmann et al. (2026)
Fraction density <-> algebraic equivalence

as *network*



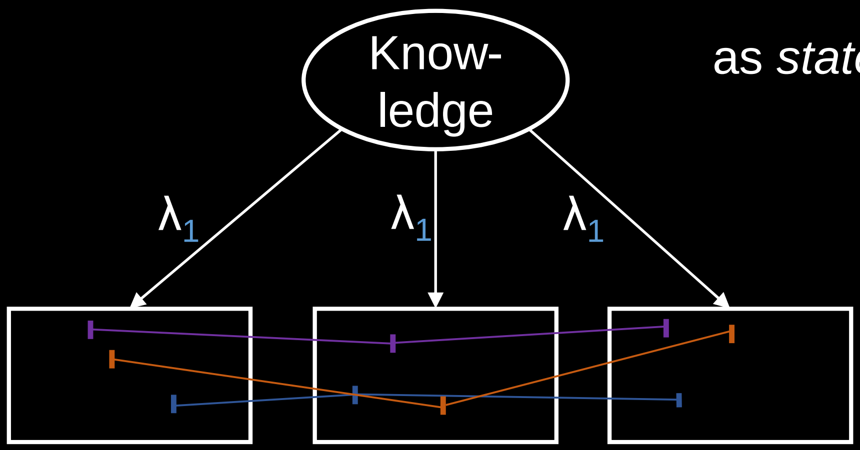
Theory

~~If responses do **not correlate** between items, you should not regard them as a **unitary construct**~~

~~0. Sum/mean scores make the assumption of equal factor loadings~~

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model



as *states*

e.g., Kleickmann, 2011; Schneider & Hardy, 2013; Tasuoka, 1985; Wilson, 1989

Theory

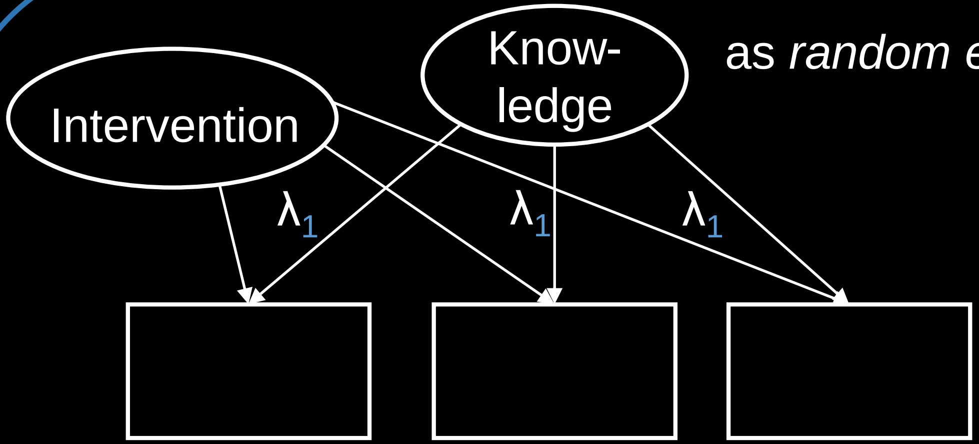
~~If responses do **not correlate** between items, you should not regard them as a **unitary construct**~~

~~0. Sum/mean scores make the assumption of equal factor loadings~~

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

as random effects



Multilevel modeling

- Items are treated as construct-relevant contexts (Donnellan et al., 2023)
- Effects of interventions or development may systematically **differ** across items
- Allows theoretical **generalization** across items (Yarkoni, 2020)

Theory

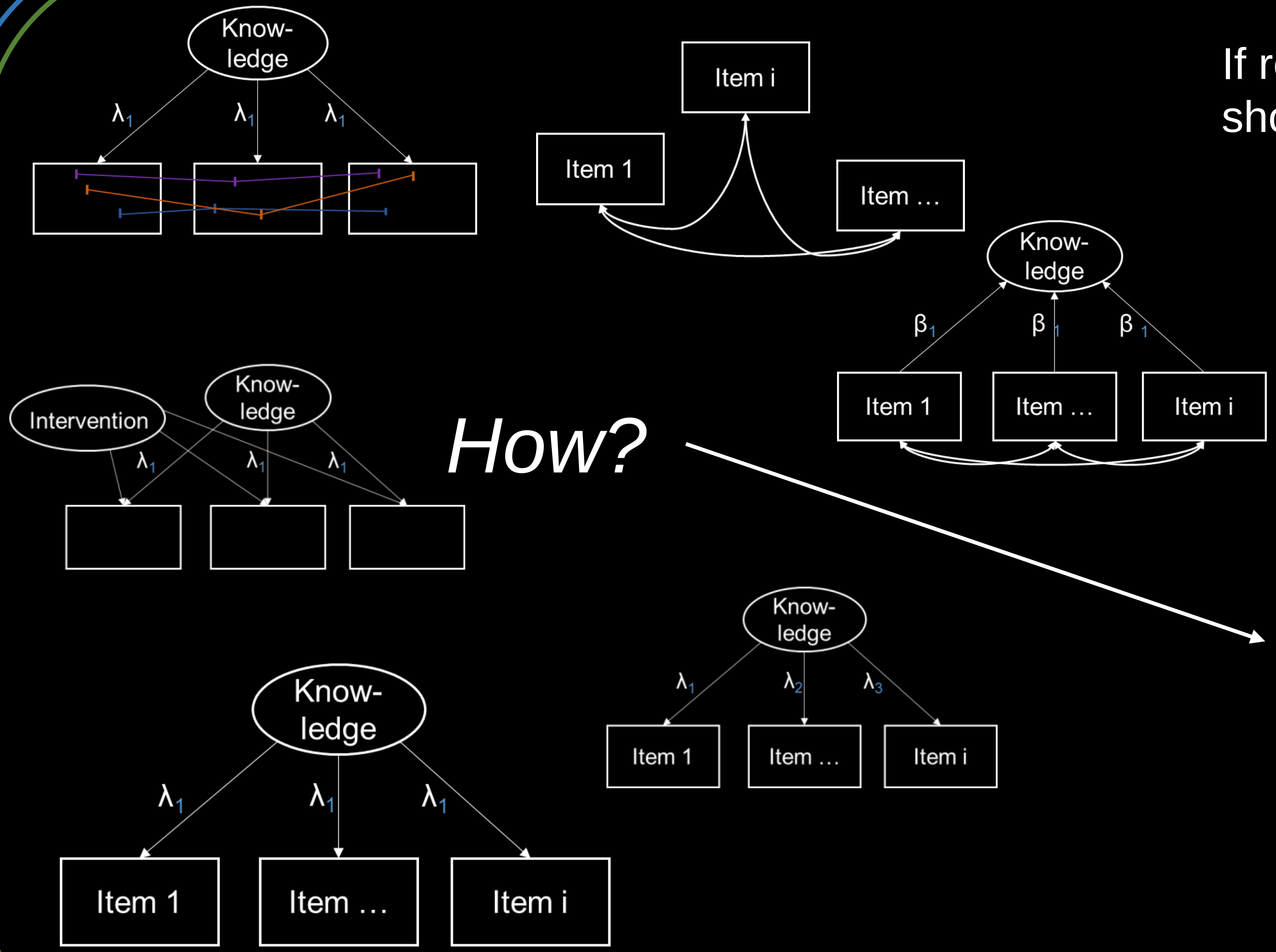
~~If responses do **not correlate** between items, you should not regard them as a **unitary construct**~~

~~0. Sum/mean scores make the assumption of equal factor loadings~~

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

Theory



~~If responses do **not correlate** between items, you should not regard them as a **unitary construct**~~

~~0. Sum/mean scores make the assumption of equal factor loadings~~

~~Group/time comparisons require measurement invariance~~

We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

Theory

Option 1: The **empirical** route

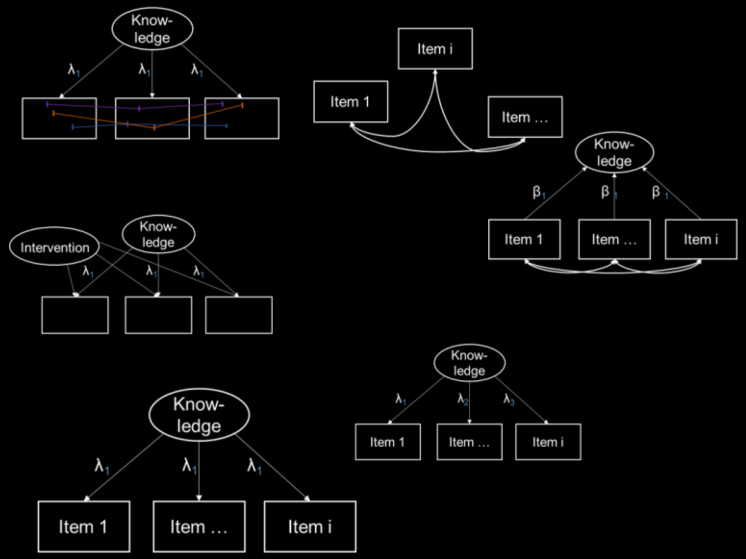
Statistical **difference tests**
between ontologies…

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

…and theories (e.g., McNeish & Wolf, 2020)



We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

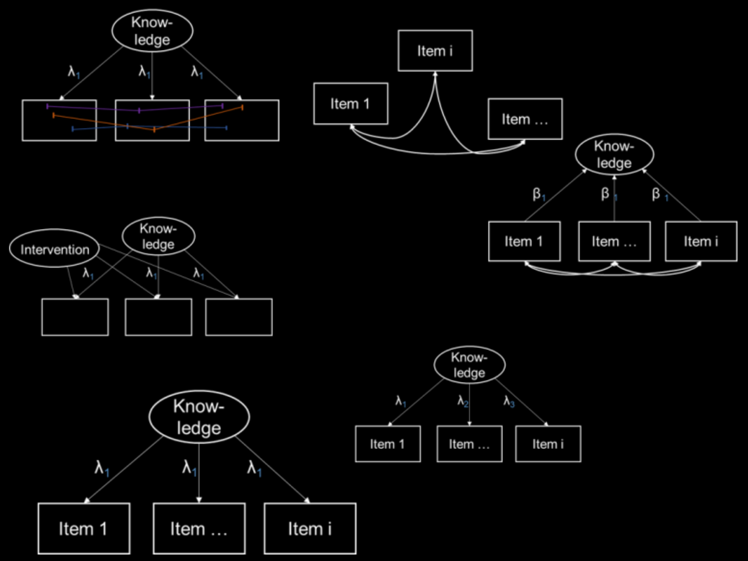
Model

Theory

Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network…

Step 2: Decide on your **theory**
General vs. item-specific effects



We can choose from different ontologies (meta-theories) and theories to describe our construct of interest

Model

Theory

One of these is **difficult**…

Option 1: The **empirical** route

Statistical **difference tests** between ontologies…

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

…and theories (e.g., McNeish & Wolf, 2020)

Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network…

Step 2: Decide on your **theory**
General vs. item-specific effects

One of these is **futile**…

Model

Theory

One of these is **difficult**…

Option 1: The **empirical** route

Statistical **difference tests** between ontologies…

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

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Option 2: The **theoretical** route

Step 1: Decide on your **ontology**
Factor vs. mixture vs. network…

Step 2: Decide on your **theory**
General vs. item-specific effects

One of these is **futile**…

Model | Theory

Beware of the **inverse fallacy** *fit ≠ theory* (Edelsbrunner, 2022)

“Factor analysis may be useful in generating hypotheses about underlying causally efficacious univariate latent variables, but does nothing to establish this” (vanderWeele, 2020)

Option 1: The **empirical** route

Statistical **difference tests** between ontologies…

Factor vs. Mixture (e.g., DeBoeck & Wilson, 2005)

Factor vs. network (e.g., van Bork et al., 2021)

Factor vs. composite (e.g., Siegel et al., 2026)

…and theories (e.g., McNeish & Wolf, 2020)

- Different ontologies typically predict the **same means, variances, covariances** (e.g., Marsman et al., 2018; Loken & Molenaar, 2007)
- Difference tests make unrealistic assumptions (**pure** ontologies)
- Theories are **causal**; assessment data **observational (**not experimental/“causal“);

Hidden causal modeling misattribution

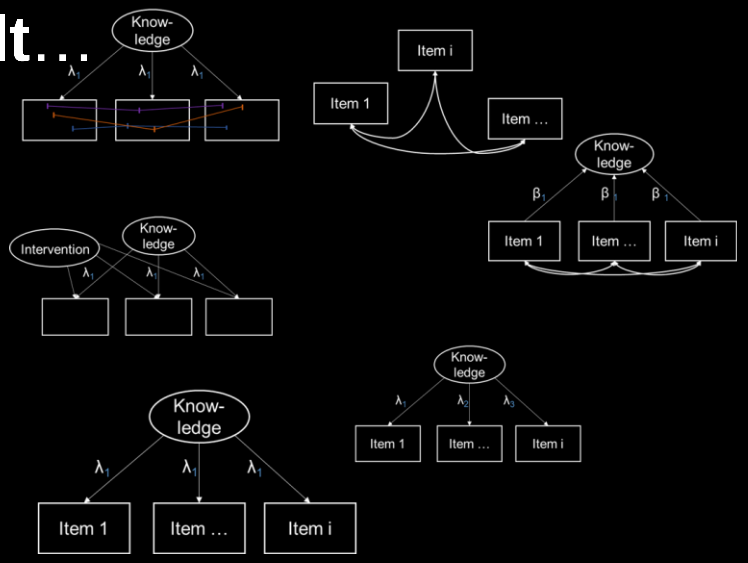
One of these is **futile**…

Model

Theory

One of these is **difficult**…

Option 2: The **theoretical** route



Step 1: Decide on your ontology
Is your construct…
…continuous (more vs. less; e.g., factor/composite)
…categorical (qualitative differences; e.g., mixture)
…both (e.g., factor-mixture, mixture-network, hybrid SEM)?

Step 2: Decide on your theory
Are your items…
…construct relevant (all contributing; composite/ML)
…construct irrelevant (all capturing the same *thing*; factor)
…both (e.g., composite/reflective hybrid SEM)?

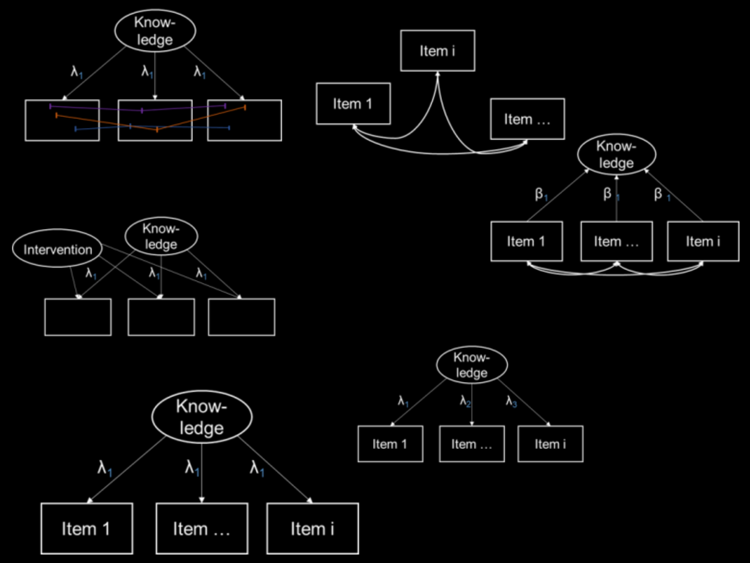
Step 3: What do you want?
…equal weights (e.g., sum score, Rasch)
…different weights (e.g., composite, factor)
…both (e.g., multilevel modeling)?

*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Model

Theory

Option 2: The **theoretical** route



How can we get these?

Theory-informed causal, formal, cognitive, & computational modeling

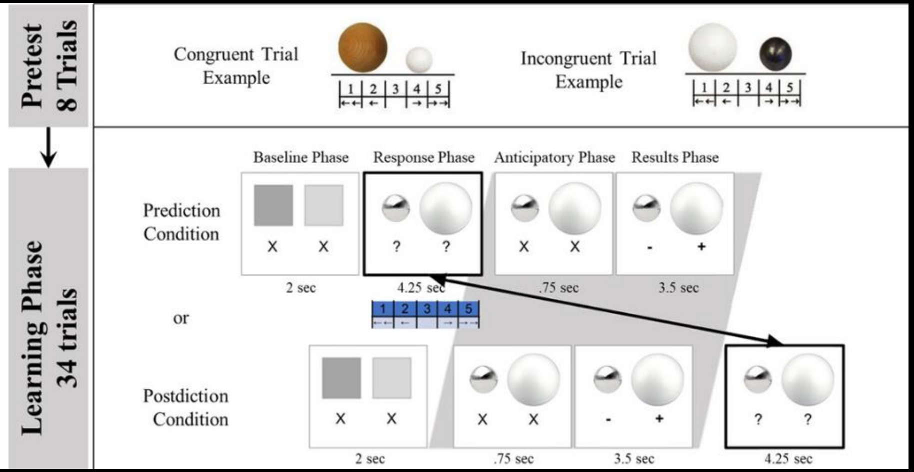
*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Model

Theory

- Colantonio, … Brod, et al. (2024)

Bayesian computational models of belief updating about water displacement «volume matters» vs. «material matters»



Knowledge tracing showing how **predicting** affects **belief revision**

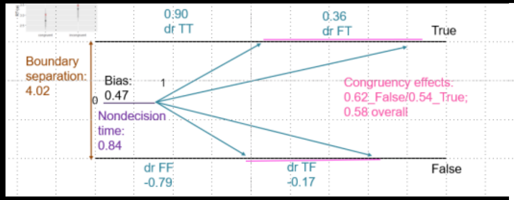
*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

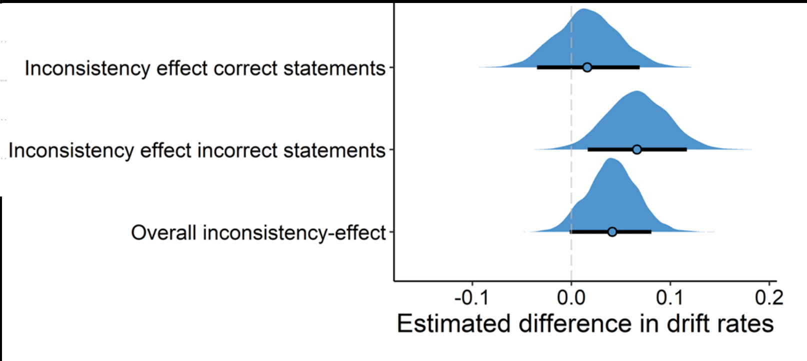
Model Theory

- Edelsbrunner & Singmann (2026)

Wiener Diffusion model about suppressing intuitive conceptions (Shtulman & Valcarcel, 2012)

«fires provide heat» vs. «coats provide heat»





We can choose from different ontologies (meta-theories) to describe our construct of interest under ***theoretical justifications***

Science expertise affects primarily **correct statements**

Longitudinal design:

No relation with inhibition

Model

Theory

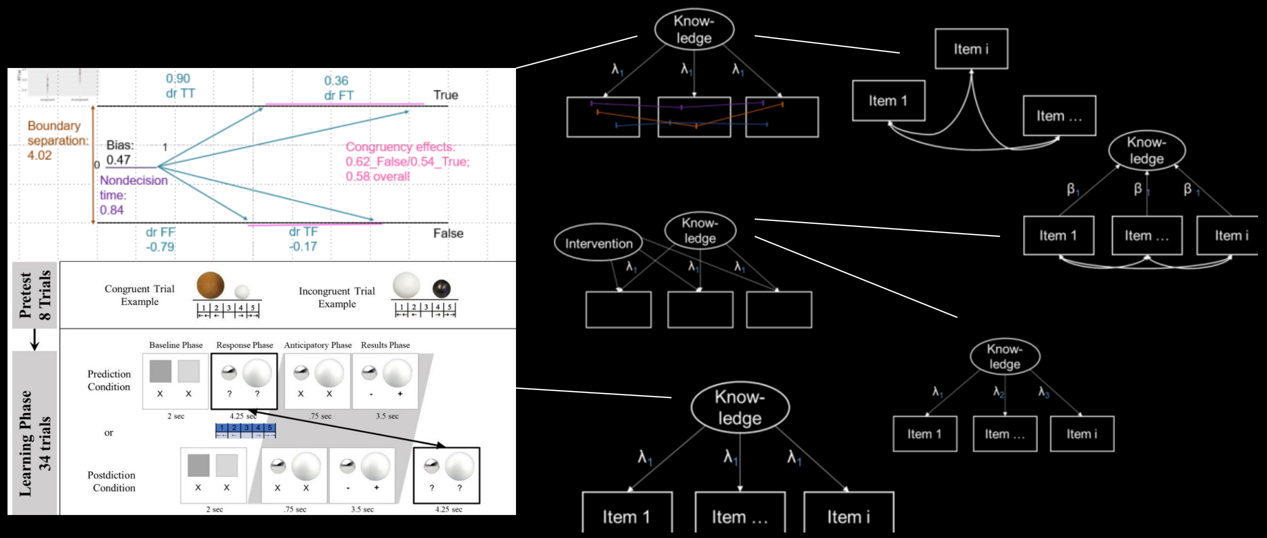
- Back to (semi-) confirmatory modeling (e.g., Wilson, 1989)
- Cognitive modeling
- Computational modeling
- Formal modeling
- Causal inference

*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Model

Theory

- The future is **cognitive**, **formalized**, **computational**, **hierarchical**, and **causally** informed (Robinaugh et al., 2021)



*We can choose from different ontologies (meta-theories) to describe our construct of interest under **theoretical justifications***

Specific wishes

- Multilevel modeling for hybrid overall/item-specific effects (Donnellan et al., 2023)

- Cognitive SEM hybrids (e.g., Rouder & Haaf, 2019)

- Data-/GenAI-driven modeling (e.g., observational learning analytics) will be **futile for causal aims** (e.g., personalization)

- Re-thinking of our field as a **cognitive science** that goes **from the process of the individual to personalized learning and population development**
- **A daring cognitive science**

Model

Theory

- We need to be **daring**

- Translate even **vague**/ad hoc **theories** into statistical/formal models

- **Reject methodological rigidity** that discounts our theories

- Emphasize and educate **interdisciplinarity** (education, cognitive science/psychology, statistics/methodology, philosophy)

**Daring, theory-driven,
& interdisciplinary modeling**

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thank you.

https://github.com/peter1328/presentations